

XRAYHEAD

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Figure 1: XRAYHEAD

ABSTRACT

XRAYHEAD creates the striking illusion of seeing a skeleton inside one's head when the experimenter touches the interior skeleton. This illusion is tested in a dark room using a setup in which the participant views a specially designed smart skeleton through a half mirror, and the experimenter touches the participant and the skeleton's head with both hands. In this specific arrangement, the distance between the skeleton head and the palm of the experimenter's hand determines how the light intensity of the LED embedded in the skeleton head increases. Alternatively, the whole part of the participant's head is kept illuminated. Additionally, to enhance illusory tactile sensation to one's skeleton, an auditory stimulus is supplied through a speaker or a bone-conduction headphone and/or modulated in sync with the degree of translucence. This creates the illusion of a head with a clearly viewed skeleton inside its translucent surface, allowing the physical touch to the skeleton's surface. This technology was tested in our laboratory exhibition, and 77 of 101 participants reported a strong tactile sensation to the illusory skeleton inside their heads.

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1 BACKGROUND

The visual expression of a transparent or translucent body is commonly shown in various visual and cultural expressions, such as

graphics, artwork, cartoons, and films. Particularly, near-future fiction (such as the optical camouflage in the animated film *Ghost in the Shell*) frequently depicts the transformation of a physical body into a transparent or translucent body. Additionally, recent psychological studies have shown that distorting body ownership using multisensory correlations can create an illusion, making the observer feel that their body comprises the transparent body part [Guterstam et al. 2013]. As a result, although the situation is fictitious, a transparent/translucent body image is fairly cognition friendly. Since it has been shown that visualizing the patient's anatomy during invasive surgery is effective for medical AR training and educational applications, this virtual reality technology of making the body transparent or translucent will likely contribute to not only the field of entertainment but also clinical treatment [Bichlmeier et al. 2007].

2 SETUP OF XRAYHEAD

2.1 XRAYSCOPE

In 2020, we designed a new experimental setup that enables users to see bone-like objects in a translucent hand by directing a smartphone's light in a specific direction, exhibited in a dark room. We called this illusion the XRAYSCOPE, and it was selected among the top 10 in the 2020 Best Illusion of the Year Contest [Imai and Kodaka 2020]. This setup uses visual mirror feedback, a classical experimental paradigm in cognitive psychology, where the participant places both arms in front of and behind a standing mirror. In XRAYSCOPE, the mirror is replaced with a half mirror, and the hand in front of the mirror is replaced with bone-like objects. With this replacement, the well-known Pepper's ghost effect is successfully applied to the space of the hidden hand. The XRAYSCOPE is completed by appropriately mediating the light intensity of the hidden space. The results of the open-laboratory experiment showed 115 of 141 participants stated that the illusion of owning the translucent hand was strong or very strong, thus proving that the illusory effect of XRAYSCOPE is consistently strong.

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2.2 XRAYHEAD

Applying the fundamentals of XRAYSCOPE to the head part of the body under the dark room condition results in the newly proposed XRAYHEAD. The system comprises two persons (one participant and one experimenter) and four pieces of equipment: a half mirror, smart skeleton head with an embedded sensor and variable-intensity LED, fixed-intensity LED to illuminate the participant, and speaker/bone-conduction headphone (Figure 2).

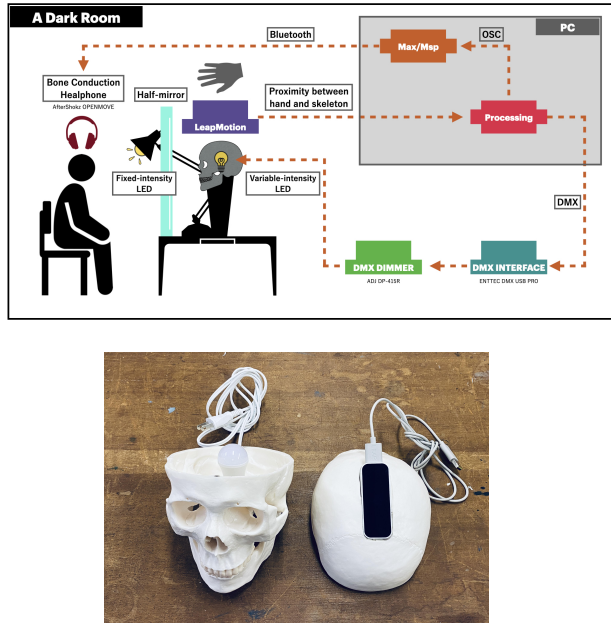


Figure 2: Detail of XRAYHEAD system

The half mirror is placed in front of the participant while sitting at a desk, and the skeleton head is placed behind the mirror. Thus, the two types of heads face each other through the half mirror. The half mirror serves as a simple mirror reflecting the participant's face as long as the hidden skeleton head does not shine since the fixed-intensity LED always illuminates the participant with sufficient intensity. Once the experimenter holds the palm of their hand against the skeleton head in the hidden space, the face viewed in the mirror begins to fade into translucence and the illuminated skeleton is replaced with the participant's face from the participant's view. This is because the light intensity of the embedded LED in the skeleton head is designed to increase proportionally to the degree of proximity between the top of the skeleton head and the palm of the experimenter's hand, which is monitored by a leap motion controller embedded on the top of the skeleton. Notably, the brightness of the skeleton head becomes maximal when the experimenter touches the skeleton's top surface. The experimenter gives this physical touch to the skeleton head in sync with a direct touch to the participant's head with the other hand. Additionally, the participant receives an auditory stimulus (generated by Max/Msp) through a speaker and/or the bone-conduction headphone, where the sound is modulated in sync with the degree of

the mirror's translucence, contributing to enhancing an illusory tactile sensation to the participant's bone.

The whole system allows the yield of the illusory visuotactile sensation where the seen skeleton head belongs to the participant, and the experimenter touches their skeleton through the invasion of the translucent surface of the head.

Although the current system does not adapt to the head motion of the participant, it is open to future improvement using some head-tracking technologies. In this assumed system, the skeleton head should be designed to rotate physically according to the participant's head rotation or it would be replaced with an XR display functioning as a "smart mirror" where the inclination of the skeleton head dynamically changes, just like the mirror.

3 REACTIONS TO THE EXHIBITION

Among the demonstrations in our open-laboratory exhibition, this technology was tested in a public experiment in February 2022 at the NTT InterCommunication Center (Shinjuku, Tokyo). We asked each participant to rate the illusion strength of experiencing two types of skeleton-head ownership, "I felt as if I saw my own skeleton through my transparent head" (visual ownership) and "I felt as if my own skeleton-head was directly touched" (tactile ownership), after each 2- or 3-min trial with XRAYHEAD using a seven-point Likert scale (from -3 to +3).

Figure 3 shows the results, which show that 70 of 101 participants (69%) agreed with visual ownership strongly (rated +2 or more), and 77 of 101 participants agreed with tactile ownership. Notably, only one participant negatively rated the illusion of tactile ownership (-1 or less than -1). Generally, the technology was found to give a consistent illusion of seeing and touching the interior skeleton of one's translucent head.

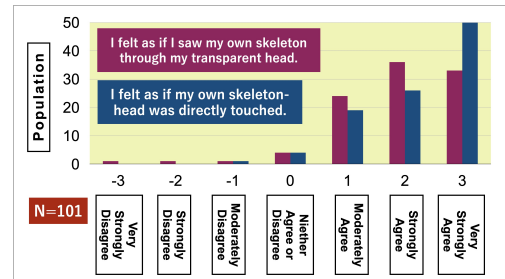


Figure 3: Evaluation of XRAYHEAD

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